Stakeholder Needs and Requirements

Author:	**Dhamdhawach Horsuwan**	**Document ID:**	MARS-REQ-001 (v1.1)
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Multi-Agent Robotic Systems (MARS) research requires a mobile robot platform capable of supporting experiments involving multiple autonomous agents operating and interacting within a shared environment. Based on practical experience operating a four-robot team, this specification identifies key stakeholders, defines priority tiers, and distinguishes stakeholder Research Needs (RN-01 to RN-08) from derived Stakeholder Requirements (SR-01 to SR-08).

1. Identified Stakeholders

ID	Stakeholder Group	Primary Role & Engineering Interest
SH-01	**MARS Researchers**	Formulate hypotheses, conduct multi-agent experiments, and publish reproducible findings.
SH-02	**Robotics & Hardware Developers**	Design, fabricate, extend, and integrate mechanical, power, sensor, and actuator modules.
SH-03	**Software & Algorithm Developers**	Implement and test decentralized control, perception, estimation, and communication software.
SH-04	**Experimental Operators**	Configure, deploy, monitor, and supervise robot fleets safely during experimental runs.
SH-05	**Laboratory Technicians**	Perform fleet charging, battery replacement, calibration, maintenance, and repairs.
SH-06	**External Research Collaborators**	Reproduce experiments, share modules, and adopt or adapt the platform across laboratories.

2. Stakeholder Priority Classification

Essential	Fundamental to core multi-agent experimentation; mandatory for the baseline platform.
Important	Significantly enhances experimental versatility and fleet efficiency; strongly targeted for inclusion.
Desirable	Adds advanced interaction, convenience, or cross-lab transferability within cost/complexity margins.

3. Research Needs (RN) and Derived Stakeholder Requirements (SR)

1. Scalability of Multi-Agent Experiments

Stakeholders: SH-01, SH-02, SH-04 — **Essential**

RN-01 Need	Researchers need to conduct experiments involving different numbers of autonomous robots to investigate how fleet size affects coordination performance, communication, task allocation, and overall multi-agent system behavior without prohibitive per-unit costs.
SR-01 Req	The platform shall support scalable multi-robot deployments across varying fleet sizes and maintain a cost-effective unit design that enables the acquisition and operation of larger robot teams.

2. Decentralized Coordination and Communication

Stakeholders: SH-01, SH-03 — **Essential**

RN-02 Need	Researchers and algorithm developers need to investigate decentralized multi-agent architectures in which robots exchange information directly and make autonomous decisions under dynamic network topologies without relying on a central server.
SR-02 Req	The platform shall provide onboard computation and peer-to-peer wireless communication interfaces that support decentralized decision-making and dynamic network topologies without requiring a centralized control server.

3. Consistent and Predictable Robot Behavior

Stakeholders: SH-01, SH-03, SH-05 — **Essential**

RN-03 Need	Researchers need individual robots within a team to exhibit consistent locomotion, sensing, and state estimation so that experimental outcomes reflect algorithmic performance rather than hardware-induced unit-to-unit variations.
SR-03 Req	The platform shall deliver repeatable and uniform motor response, odometry, localization accuracy, and sensor performance across all robot units in the fleet.

4. Adaptability to Different Research Problems

Stakeholders: SH-01, SH-02, SH-03 — **Important**

RN-04 Need	Researchers and developers need to evaluate diverse algorithms, sensors, actuators, and experimental payloads across multiple projects without redesigning the core robot platform for each study.
SR-04 Req	The platform shall feature modular mechanical mounting, standardized electrical/communication interfaces, and extensible software architecture to allow adding or swapping sensors, actuators, and compute modules without full-system redesign.

Stakeholder Needs and Requirements

3. Research Needs (RN) and Derived Stakeholder Requirements (SR) (Continued)

5. Reliable Long-Term and Fleet-Level Operation

Stakeholders: SH-01, SH-04, SH-05 — **Essential**

RN-05 Need	Experimental operators and technicians need to run multiple robots concurrently over extended experimental sessions with minimal downtime for recharging, battery swaps, routine maintenance, or fault recovery.
SR-05 Req	The platform shall incorporate reliable power management, integrated battery protection, accessible battery replacement, efficient fleet charging, and modular serviceability to minimize operational downtime as fleet size scales.

6. Environmental Awareness and Physical Interaction

Stakeholders: SH-01, SH-02, SH-03 — **Important / Desirable**

RN-06 Need	Researchers need robots to perceive and navigate cluttered shared environments safely, avoid inter-robot and obstacle collisions, and-when required by specific studies-physically interact with objects or perform cooperative transport.
SR-06 Req	The platform shall provide onboard environmental sensing for navigation and collision avoidance (Important), and shall accommodate optional mechanical attachments or payloads for physical interaction and cooperative object transport (Desirable).

7. Experimental Monitoring and Data Collection

Stakeholders: SH-01, SH-03, SH-04 — **Important**

RN-07 Need	Researchers and operators need to observe, record, and analyze the real-time state and historical behavior of both individual robots and the collective fleet without interfering with the robots' decentralized autonomy.
SR-07 Req	The platform shall support non-intrusive centralized fleet monitoring, real-time telemetry streaming, synchronized experimental data logging, and visualization tools while preserving decentralized onboard execution.

8. Usability, Reproducibility, and Technology Transfer

Stakeholders: SH-01, SH-04, SH-05, SH-06 — **Desirable**

RN-08 Need	Researchers, students, and external collaborators from varied engineering and software backgrounds need to onboard quickly, reproduce experimental setups reliably, and replicate or adopt the platform across different research groups.
SR-08 Req	The platform shall provide intuitive user interfaces, standardized configuration workflows, built-in diagnostic indicators, and comprehensive documentation to reduce the learning curve and facilitate reproducibility and technology transfer.

4. Revision History & Traceability Log

v1.1 (2026-11-12)	Separated Research Needs (RN-01..08) & Stakeholder Requirements (SR-01..08); added Stakeholders & Priorities.
v1.0 (2026-11-12)	Initial baseline release defining the 8 core multi-agent robotic system needs (RN-01..08). Archived in docs/archive/.